A.2 Second Research/Programming Assignment

MSDS 458

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Lindsey Chenault

**Abstract**

This project investigates how various network design choices influence performance, generalization, and reliability in image classification tasks. More specifically, the analysis focuses on three fundamental aspects: network architecture, model depth, and regularization techniques. This project provides insight into the trade-offs that are involved in designing effective and robust neural networks by evaluating how variations in these elements affect classification outcomes. Although gaining a complete understanding requires ten experiments, this study conducted an additional experiment to compare fully connected deep neural networks (DNNs) and convolutional neural networks (CNNs) on the CIFAR-10 dataset across a range of architectural and regularization settings. Dense networks with two and three hidden layers consistently achieved low classification accuracy, which means that it is difficult for fully connected architectures to capture the spatial structure that is present in image data. An additional dense model that incorporated dropout and L2 regularization further showed that regularization by itself is insufficient to overcome these limitations in architecture.

On the other hand, CNN models that use convolutional and pooling layers learned localized visual features like edges, textures, and shapes, which ultimately led to substantially higher classification accuracies. Deeper CNN architectures showed better generalization than models that were shallower, although networks that were trained without regularization displayed evident signs of overfitting. Regularization was important in improving the models' reliability, with dropout consistently reducing variance and improving test performance (Srivastava et al., 2014). Batch normalization improved training stability but did not sufficiently address overfitting when used alone (Ioffe & Szegedy, 2015). Among the approaches evaluated in this study, CNNs that combined dropout and batch normalization delivered the most durable

and reliable performance. Overall, these findings mean that effective computer vision models benefit from convolutional architectures that are combined with proper regularization to achieve both high accuracy and reliable generalization (Zhao et al., 2024).

## Introduction

Image classification is very important in the modern world, as is technology like facial recognition, autonomous systems, mobile device security, and medical imaging. As these systems deploy more widely, their decisions carry weight and can have consequences, which makes accuracy, reliability, and generalization all the more important. While deep learning models frequently perform well, not all neural network architectures are equally effective or trustworthy when applied to image data (Zhao et al., 2024).

From the point of view of a business, models that perform well during training but fail to generalize can lead to high costs, which include diminished trust from users and increased safety or security risks. For example, misclassification in facial recognition systems may give someone unauthorized access or fail to authenticate valid users, while errors in medical imaging can delay a diagnosis or treatment. Given these risks, there is a need for models that are not only accurate but also stable when they are exposed to new data.

Dense neural networks provide a useful baseline because of how simple they are, and there are lower computational requirements. However, they treat image pixels as independent inputs and ignore spatial relationships. This restriction makes them poorly suited for visual data (LeCun et al., 1998). In contrast, convolutional neural networks exploit spatial locality through convolutional and pooling operations, enabling hierarchical feature learning that better aligns with image structure (LeCun et al., 1998; Zhao et al., 2024).

Outside of architecture, model depth and regularization are important when considering controlling overfitting and improving generalization. While deeper models increase representational capacity, they can also present instability and higher computational cost (Krizhevsky et al., 2012). Regularization techniques like dropout, batch normalization, and L2 weight decay are typically used to address these challenges that arise, though their effectiveness depends on implementation and architectural design (Srivastava et al., 2014; Ioffe & Szegedy, 2015). This investigation focuses on controlled comparisons to evaluate how architecture, depth, and regularization jointly influence performance and reliability in image classification, with implications for both technical evaluation and decisions from a managerial standpoint.

## Literature Review

Convolutional neural networks (CNNs) have become the dominant approach in computer vision due to their ability to learn meaningful patterns directly from raw image data. In a comprehensive review, Zhao et al. (2024) analyze CNN architectures and core design principles, which offer understanding into how architectural and regularization choices form model performance and generalization. Their work sets CNNs as an unmistakable advancement over traditional machine learning methods for image-based tasks.

According to Zhao et al. (2024), the effectiveness of CNNs stems from fundamental structural properties like local connectivity, weight sharing, and hierarchical feature learning. Through convolutional and pooling layers, CNNs capture spatial relationships within images, with early layers learning simple features such as edges and textures, and deeper layers extracting more conceptual representations. This hierarchical feature learning process was foundationally founded by LeCun et al. (1998) and is still prominent to modern computer vision

systems. Previous work has also shown that increasing network depth can improve classification accuracy but raises optimization challenges and a greater risk of overfitting if not adequately managed (Krizhevsky et al., 2012). This shows how important it is to evaluate architectural trade-offs rather than assume that deeper networks are intrinsically superior.

Researchers have widely studied regularization techniques like dropout and batch normalization as methods for improving generalization and training stability. Dropout lowers reliance on individual neurons and improves generalization (Srivastava et al., 2014), while batch normalization stabilizes training by normalizing intermediate activations (Ioffe & Szegedy, 2015). However, much of the current literature focuses on high-performing CNNs and provides limited direct comparisons with dense neural networks under equivalent experimental conditions (Zhao et al., 2024).

Researchers often exclude dense neural networks as significant baselines in computer vision research. Although dense models are known to perform poorly on image classification tasks, fewer studies explicitly demonstrate why regularization by itself cannot compensate for architectural limitations. By including heavily regularized dense models alongside CNNs, this study provides empirical evidence that architectural suitability plays a more critical role than regularization strength alone, consistent with the framework outlined by Zhao et al. (2024).

**Methods**

A structured sequence of experiments was run to assess how neural network architecture, depth, and regularization affect classification performance, generalization, and training behavior through the use of dense neural networks and CNNs under controlled conditions. This study assesses dense neural networks and convolutional neural networks under controlled conditions

by systematically varying architectural and regularization choices. Architectural design and regularization methods were adjusted across experiments, while the dataset, optimizer, batch size, and training protocol were held constant to guarantee fair and significant comparisons across models.

All models were implemented using TensorFlow and Keras. Dense neural networks consisted of two or three fully connected hidden layers with ReLU activation functions, then followed by a softmax output layer. Convolutional neural networks used two or three convolution–max pooling blocks, with the number of filters increasing in deeper layers to capture higher-level features (LeCun et al., 1998). Regularized variants incorporated dropout, batch normalization, L2 weight decay, or combinations of these methods (Srivastava et al., 2014; Ioffe & Szegedy, 2015). Each model terminated with a dense classification head.

All experiments used the CIFAR-10 dataset, which contains 60,000 color images across ten classes. Images were normalized to the [0, 1] range to improve numerical stability during training. The original training set was split into training and validation subsets using a 90/10 split, with the validation set used exclusively for monitoring generalization and early stopping. No data augmentation was applied in order to isolate the effects of architectural and regularization changes.

Training was performed using the Adam optimizer with a batch size of 64 for a maximum of 200 epochs. Early stopping was applied based on validation accuracy to prevent overfitting and avoid unnecessary computation. Model checkpoints were saved, and the best-performing model was restored for final evaluation on the test set. Performance was evaluated using

accuracy and loss metrics, confusion matrices, and feature map visualizations, providing both quantitative and qualitative insight into model behavior (Zhao et al., 2024).

Assuring experimental control was a main priority in this research. All models were trained using identical optimization settings, including learning rate, batch size, and early stopping criteria. This consistency confirms that observed performance differences are attributable to architectural design and regularization rather than to confounding training variables. Without this level of control, comparisons across experiments would be difficult to interpret.

Reproducibility was also considered throughout the experimental process. Random seeds were fixed for dataset splitting and model initialization to reduce stochastic variability. While deep learning models inherently involve some randomness, these precautions improve consistency across runs and bolster the reliability of the reported results.

In addition to accuracy metrics, training time and parameter counts were recorded to consider computational efficiency. These measures are particularly relevant in applied settings where scalability and cost constraints influence model selection. A model that achieves marginally higher accuracy at a significantly higher computational cost may not be optimal for deployment, especially in environments with limited resources.

**Results**

Experiments 3 through 10 evaluate convolutional architectures at different depths and with distinct regularization strategies, while Experiment 11 delivers a heavily regularized dense-network baseline for comparison. Across 11 experiments, dense neural networks

invariably underperformed when compared to convolutional architectures. Experiments 1 and 2 showed that increasing the depth of dense networks did not improve performance and, in fact, led to poorer generalization. Experiment 5 showed that adding dropout caused underfitting, and Experiment 11 demonstrated that even when dropout and L2 regularization were combined, the dense model achieved only limited gains in test accuracy, which highlights persistent structural limitations. Overall, these findings mean that regularization by itself cannot compensate for the lack of spatial inductive bias (LeCun et al., 1998; Zhao et al., 2024).

Introducing convolutional layers in Experiment 3 led to a substantial increase in accuracy, though the gap between training and validation performance indicated overfitting. Increasing convolutional depth in Experiment 4 improved generalization and narrowed this gap, which suggests that deeper hierarchical feature extraction improves strength (Krizhevsky et al., 2012). Experiments 6 through 8 showed that dropout effectively controlled overfitting, whereas batch normalization alone predominantly improved training stability without sufficiently enhancing generalization (Srivastava et al., 2014; Ioffe & Szegedy, 2015). The best-performing model combined dropout and batch normalization, achieving the highest validation and test accuracy with minimal overfitting.

Experiments 9 and 10 demonstrated the importance of carefully tuning the strength of regularization. Moderate regularization improved training stability and reduced variance, whereas excessive regularization constrained learning capacity and degraded performance. Overall, the results indicate that CNN architecture primarily drives performance, while regularization shapes reliability and generalization behavior (Zhao et al., 2024).

Past final accuracy values and learning curves provided substantial insights into how different models behaved during training. Dense neural networks continually showed early stagnation in validation accuracy, which indicates finite capacity to learn discriminative features from image data. Even when training accuracy increased, validation performance failed to improve, which shows poor generalization rather than undertraining.

In contrast, CNN models demonstrated more stable learning dynamics. Validation accuracy increased steadily throughout training, specifically in models that incorporated dropout. This pattern means that convolutional architectures better serve the task of extracting significant features while remaining robust to unfamilair data (LeCun et al., 1998). The smaller difference between training and validation accuracy in regularized CNNs further supports this conclusion.

Experiment 11 provides a precise contrast to the CNN results. Despite incorporating both dropout and L2 regularization, the dense model still exhibited unsatisfactory generalization performance. This outcome reinforces the conclusion that regularization techniques cannot substitute for architectural alignment with the underlying data structure. Without spatial feature extraction, dense networks remain fundamentally limited in image classification tasks (Zhao et al., 2024).

**Conclusions**

This study proves just how important neural network architecture and regularization choices are in building reliable image classification models. Dense neural networks were incapable of generalizing effectively on image data, even when deep and heavily regularized. These findings reinforce the limitations of fully connected architectures for tasks involving complex spatial structure (LeCun et al., 1998; Zhao et al., 2024).

In contrast, convolutional neural networks invariably achieved superior performance by exploiting spatial locality and hierarchical feature learning. Regularization was important for controlling overfitting, with dropout emerging as the most effective technique (Srivastava et al., 2014). The blend of dropout and batch normalization had the most stable and reliable results across all experiments (Ioffe & Szegedy, 2015).

From a management and business perspective, these findings stress how important it is in choosing models that balance accuracy, generalization, and efficiency rather than maximizing any single metric. CNNs with the right regularization deliver a more dependable and cost-effective solution than larger dense models, especially in real applications where reliability and trustworthiness are crucial. Overall, this research provides practical guidance for deploying deep learning systems that perform consistently and responsibly in image classification tasks.

From a strategic perspective, these findings have immediate implications for organizations deploying machine learning systems. Investing in model complexity without architectural suitability can lead to counterproductive use of resources and unreliable outcomes. The results of this study show that selecting the appropriate model structure is more impactful than increasing depth or applying stronger regularization.

For managers and decision-makers, this research spotlights the matter of evaluating models based on stability and generalization rather than peak accuracy alone. Models that exhibit consistent validation performance and small generalization gaps are a better fit for long-term deployment. Incorporating such evaluation criteria into model selection processes can decrease operational risk and remediate system reliability.

Overall, this study affirms that adequate machine learning deployment requires alignment between data structure, model architecture, and training strategy. By systematically comparing dense and convolutional networks across multiple regularization settings, the research provides actionable direction for building reliable and efficient image classification systems.

Consider a practical computer vision task like facial recognition for user authentication on a mobile device. In this setting, the objective is not simply to maximize classification accuracy on a benchmark dataset, but to reliably distinguish an authorized user from illicit individuals under real conditions. These conditions include variations in lighting, facial expressions, camera angles, partial occlusion, and changes in appearance over time. Training such a system would require an enormous and myriad dataset of facial images, with multiple examples per user captured across different environments. The data would be divided into training, validation, and test sets to ensure that the model generalizes beyond the specific images used during training.

A facial recognition system of this nature would require deep learning rather than a shallow or single-hidden-layer network. Shallow models lack the representational capacity to capture the complex, hierarchical structure of facial features, which range from low-level edges and textures to higher-level configurations such as eyes, noses, and facial geometry. Deep neural networks enable successive layers to learn increasingly abstract features, making them right for modeling the variability inherent in faces. As demonstrated in this study, increasing depth alone is insufficient unless the architecture lines up with the data's structure (Krizhevsky et al., 2012).

Convolutional neural networks show an obvious advantage for facial recognition tasks by leveraging spatial locality and translation invariance. Convolutional layers learn localized

patterns that are mighty to small shifts and distortions, which are common in mobile-device imagery. Pooling layers also improve robustness by decreasing sensitivity to exact pixel locations. These properties let CNNs generalize more effectively than dense neural networks, which treat pixels as independent inputs and ignore spatial relationships (LeCun et al., 1998; Zhao et al., 2024). The best performance and stability of CNNs observed in this research supply verifiable support for their use in authentic facial recognition systems.

Evaluating the accuracy of a facial recognition model needs more than overall classification accuracy. In practice, practitioners need to assess performance using metrics like false acceptance rate (unauthorized access) and false rejection rate (failure to recognize an authorized user), as these errors carry different business and security implications. A reliable system must balance convenience and security, which minimizes both types of errors under realistic operating conditions. Real-user testing, including repeated authentication attempts over time, is important to make sure that the model remains robust after deployment. These considerations reinforce the importance of selecting architectures that generalize well and exhibit stable behavior, rather than models that achieve high accuracy only under controlled experimental settings.

**References**

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https://doi.org/10.1007/s10462-024-10721-6

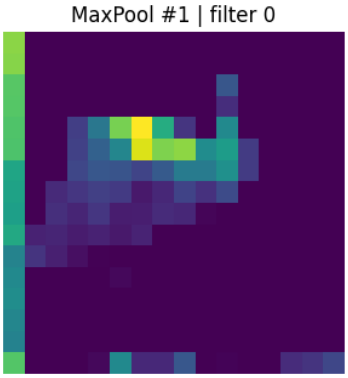
# Appendix

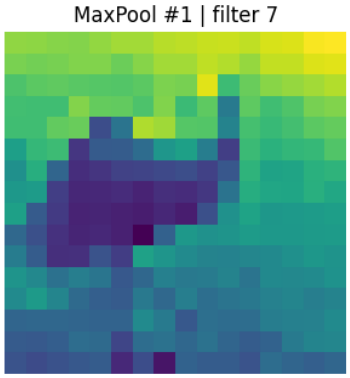
## Results 1

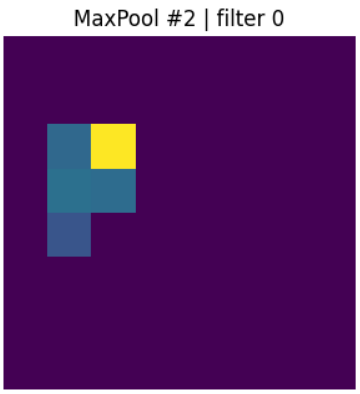
| experiment | params | train_loss | train_acc | val_loss | val_acc | test_loss | test_acc | train_time_sec | optimizer | batch_size | epochs_ran |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Exp1_DNN_2hidden_no_reg | 1255882 | 1.244229 | 0.559467 | 1.423928 | 0.5020 | 1.411031 | 0.5050 | 164.68 | adam | 64 | 17 |
| Exp2_DNN_3hidden_no_reg | 1738890 | 1.366026 | 0.509356 | 1.527360 | 0.4518 | 1.493653 | 0.4712 | 283.77 | adam | 64 | 9 |
| Exp3_CNN_2block_no_reg | 545098 | 0.229252 | 0.920511 | 1.270858 | 0.6958 | 0.887322 | 0.7057 | 1201.50 | adam | 64 | 16 |
| Exp4_CNN_3block_no_reg | 356810 | 0.316399 | 0.887667 | 0.923465 | 0.7428 | 0.794473 | 0.7432 | 1142.61 | adam | 64 | 10 |
| Exp5_DNN_2hidden_dropout | 1255882 | 1.737181 | 0.373000 | 1.706320 | 0.3936 | 1.684457 | 0.4053 | 256.20 | adam | 64 | 16 |
| Exp6_CNN_3block_dropout | 356810 | 0.449504 | 0.846000 | 0.645920 | 0.7698 | 0.682221 | 0.7651 | 2794.48 | adam | 64 | 23 |
| Exp7_CNN_3block_batchnorm | 358218 | 0.523600 | 0.819400 | 0.780100 | 0.7368 | 1.197226 | 0.6707 | 564.95 | adam | 64 | 7 |
| xp8_CNN_3block_batchnorm_dropout | 358218 | 0.477204 | 0.834933 | 0.637082 | 0.7770 | 0.654134 | 0.7735 | 2657.30 | adam | 64 | 15 |
| Exp9_CNN_2block_L2_dropout | 545098 | 0.662891 | 0.827044 | 0.899798 | 0.7258 | 0.920680 | 0.7241 | 2099.44 | adam | 64 | 19 |
| Exp10_CNN_2block_strong_reg | 545098 | 0.926563 | 0.758889 | 1.036976 | 0.7116 | 1.053396 | 0.7101 | 2556.86 | adam | 64 | 24 |

| experiment | params | train_loss | train_acc | val_loss | val_acc | test_loss | test_acc | train_time_sec | optimizer | batch_size | epochs_ran |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Exp11_DNN_3hidden_dropout_L2 | 1738890 | 1.712304 | 0.403356 | 1.750123 | 0.3886 | 1.725983 | 0.3946 | 714.75056 | adam | 64 | 20 |

## Results 2 (Experiment 3)

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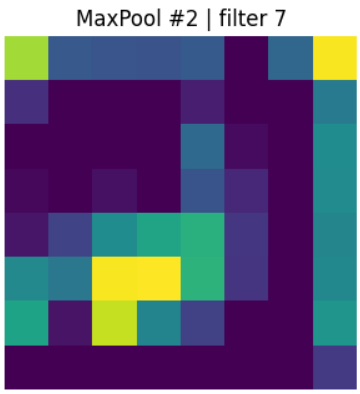
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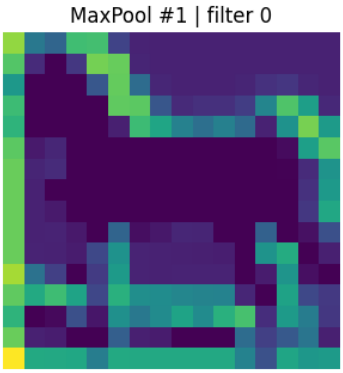
Test image idx=2004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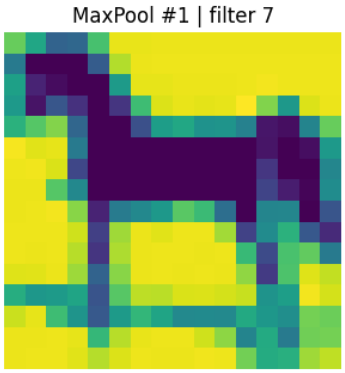
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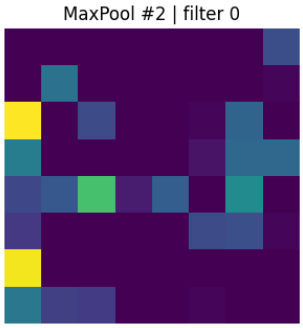
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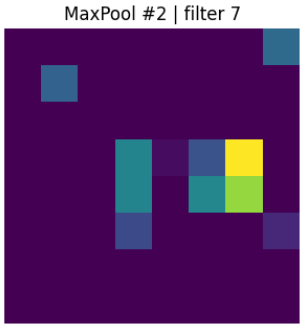
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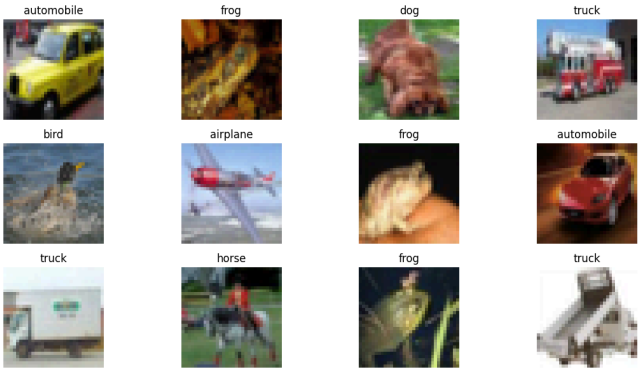
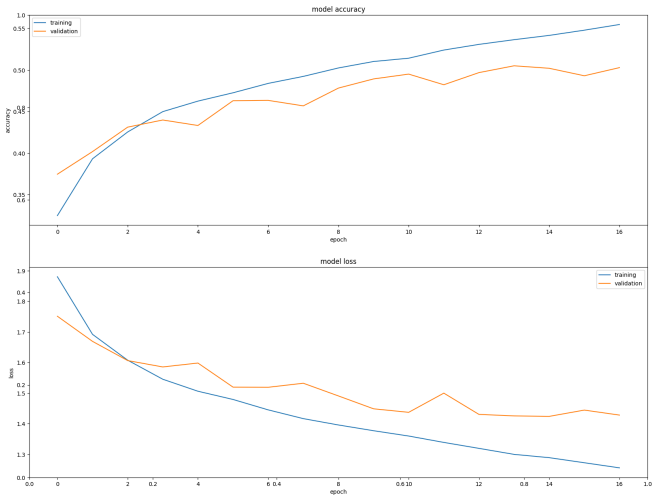
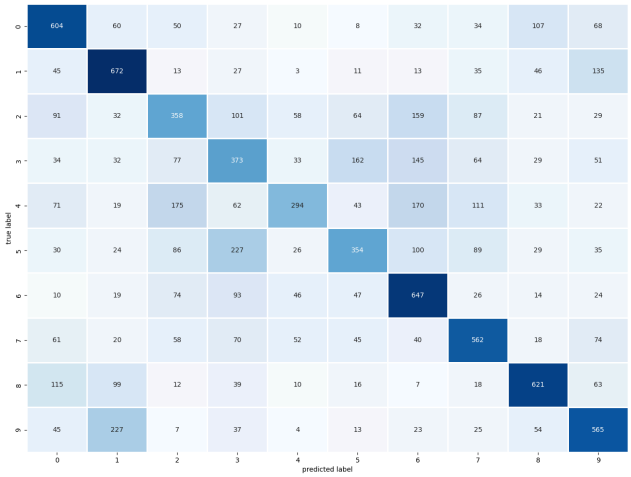

automobile
frog
dog
truck
bird
airplane
frog
automobile
truck
horse
frog
truck

# Experiment 1 (In-Depth)

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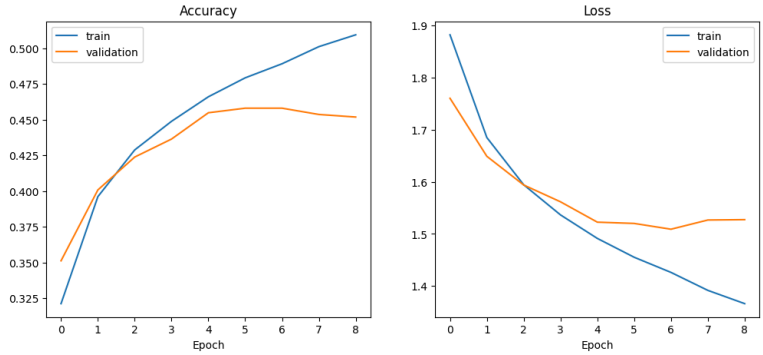


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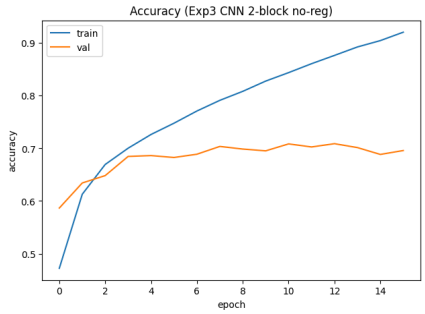
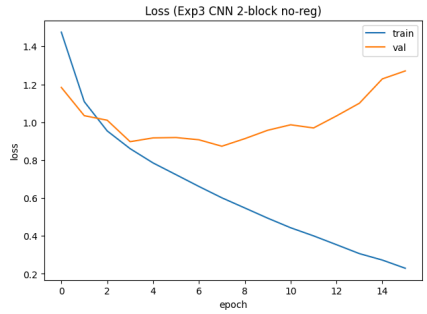
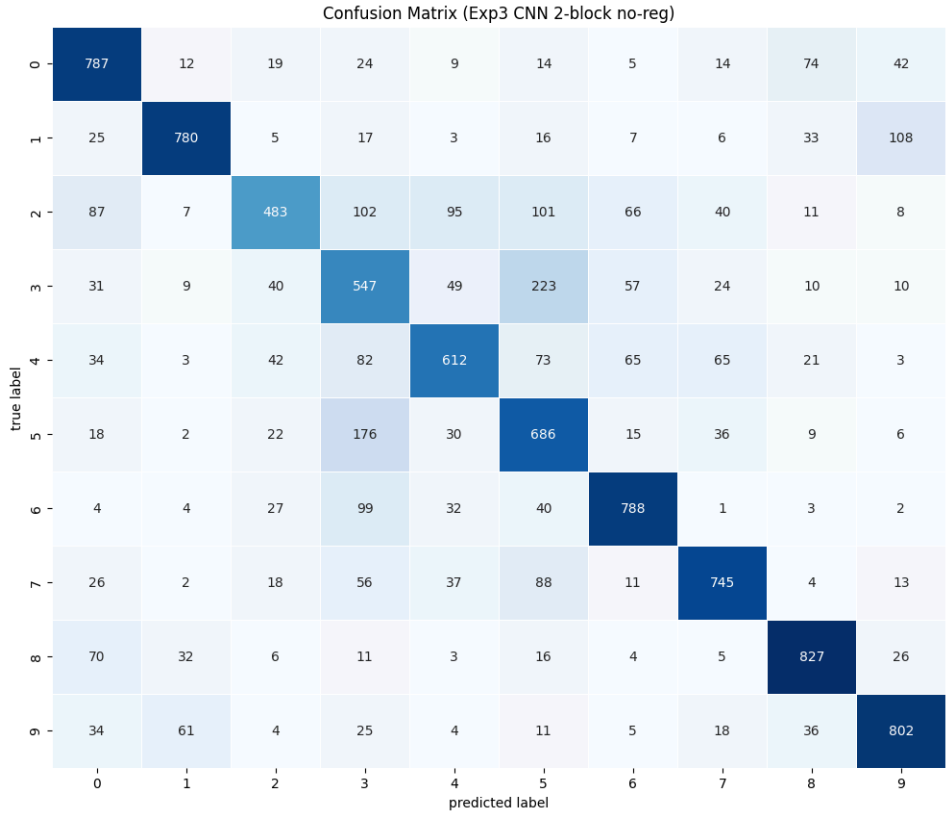
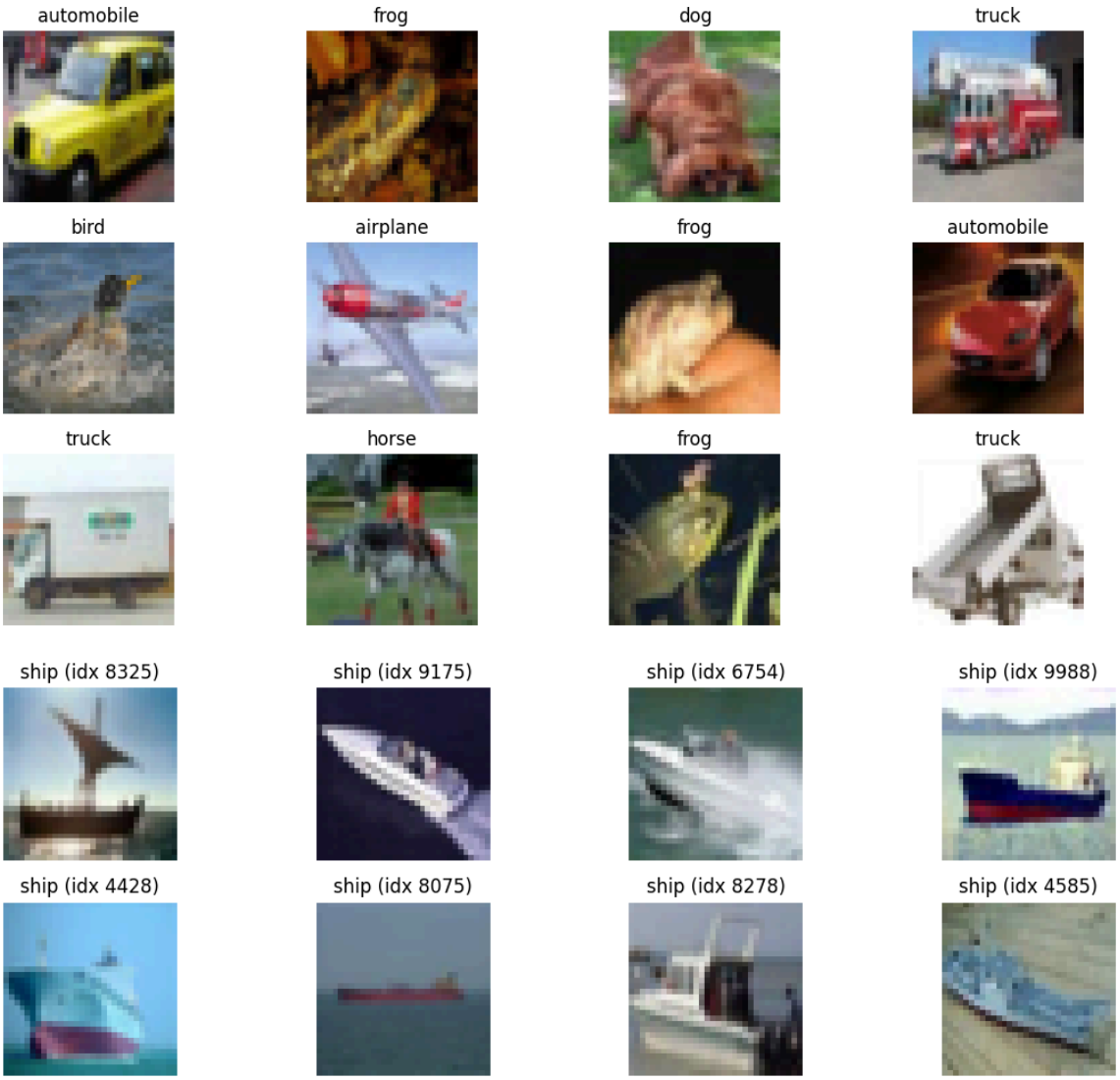

Accuracy (Exp3 CNN 2-block no-reg)
train
val
accuracy
epoch


Loss (Exp3 CNN 2-block no-reg)
train
val
loss
epoch


Confusion Matrix (Exp3 CNN 2-block no-reg)
true label
predicted label

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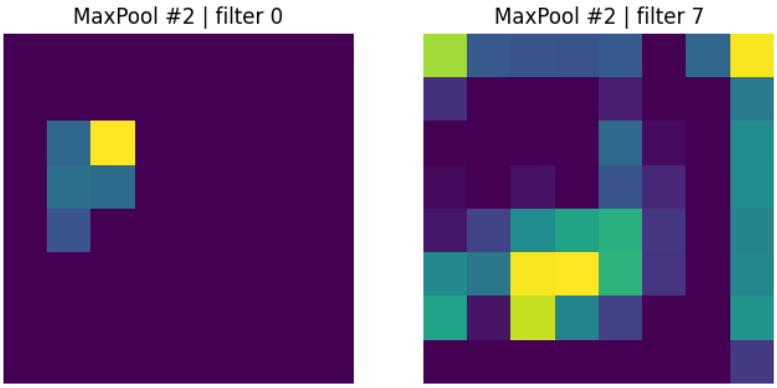


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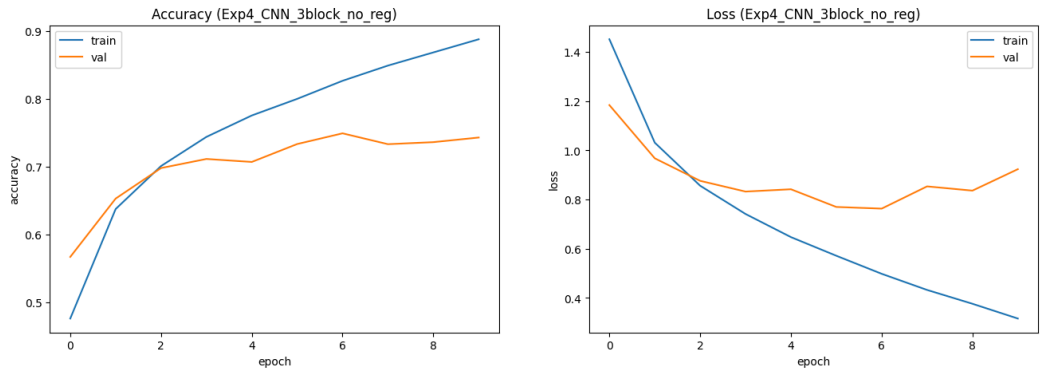


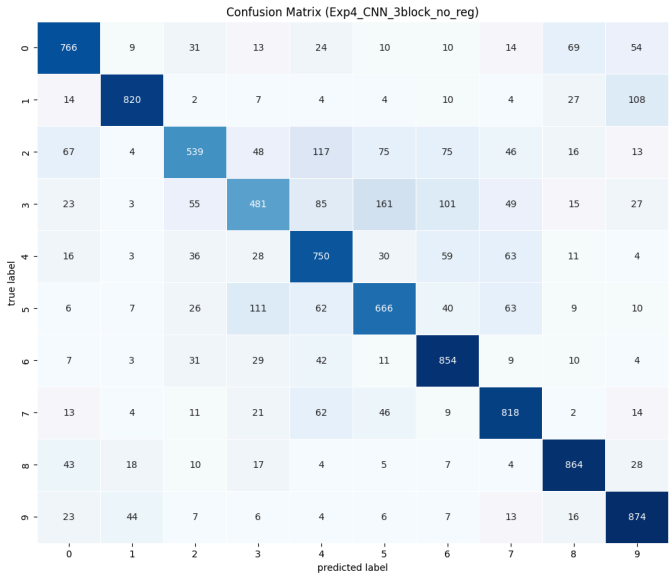

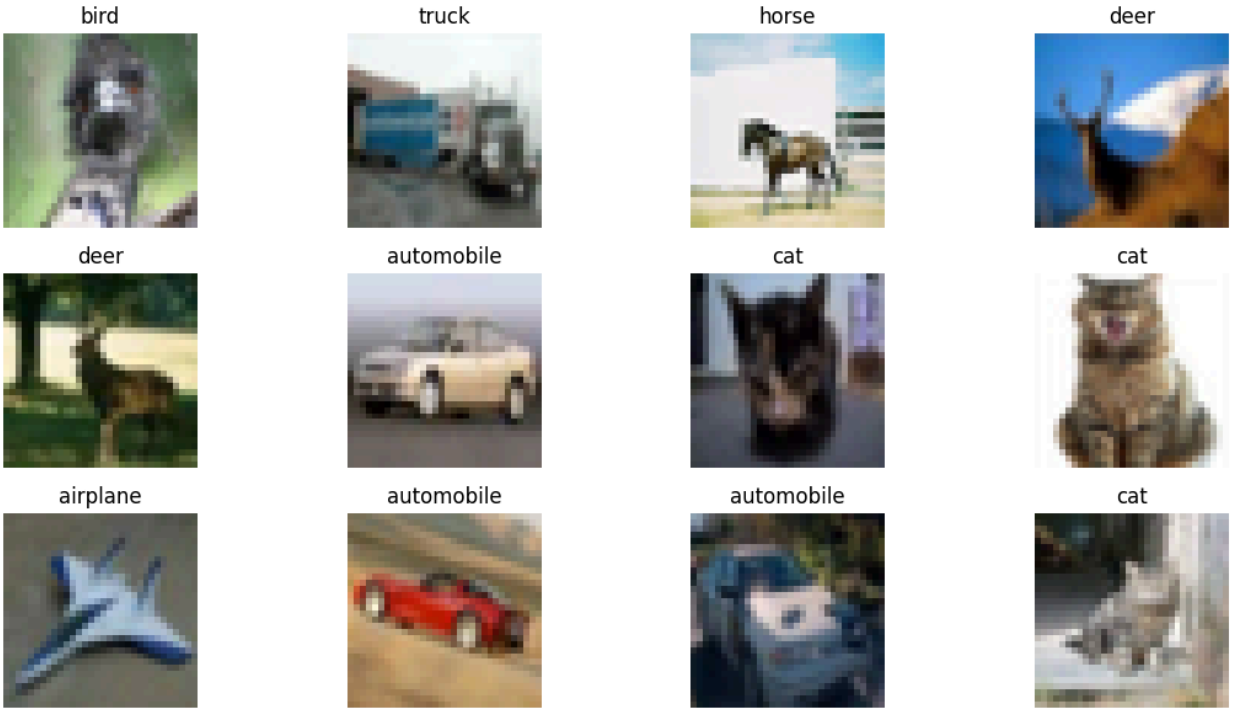


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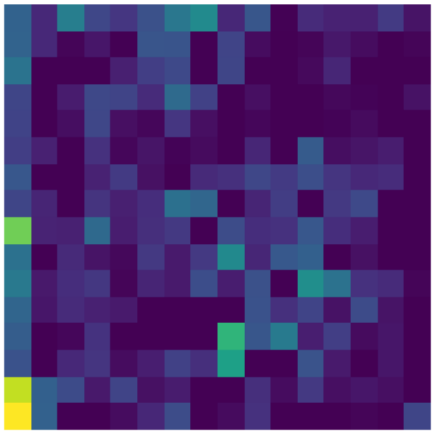
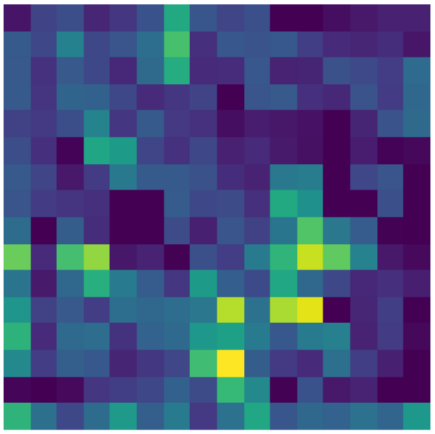
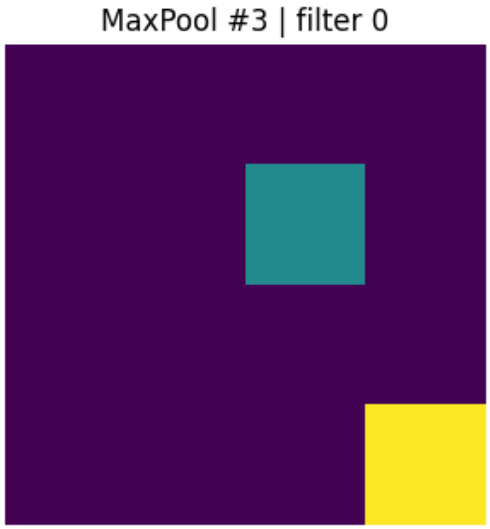
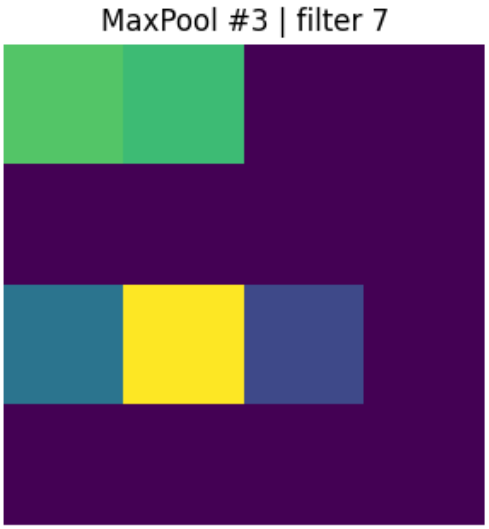


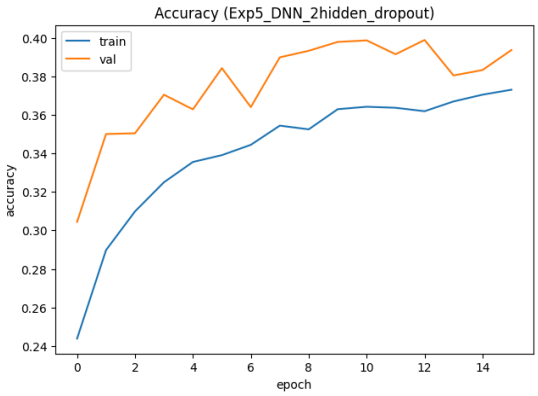
MaxPool #1 | filter 7

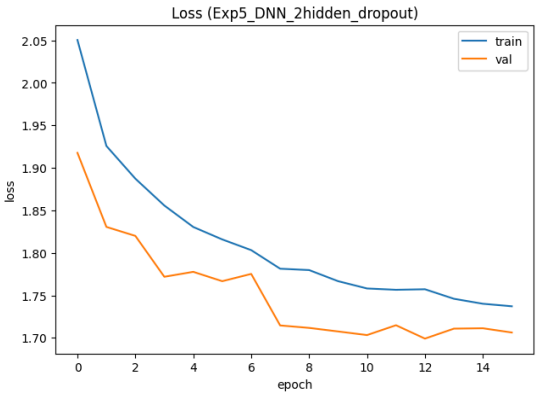


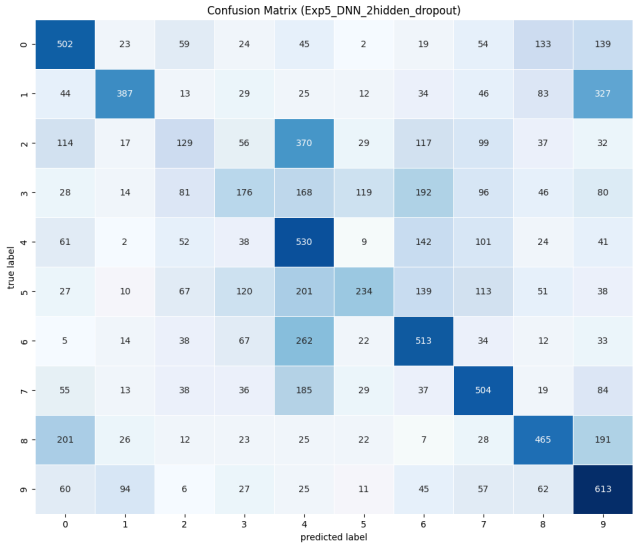


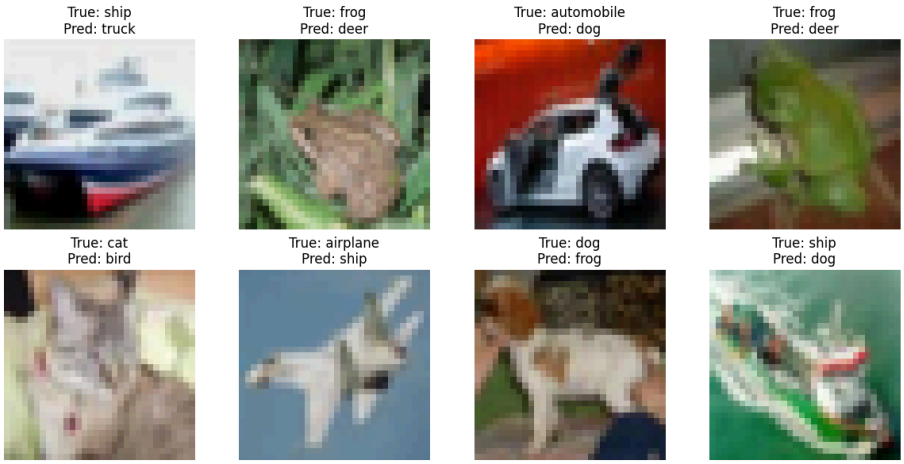




## Experiment 5 (In-Depth)

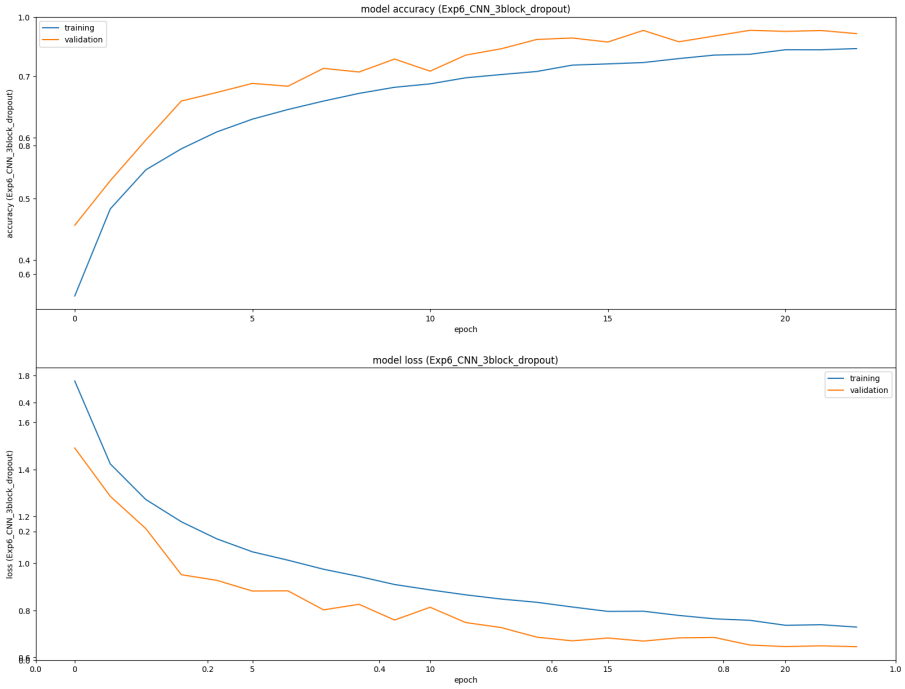


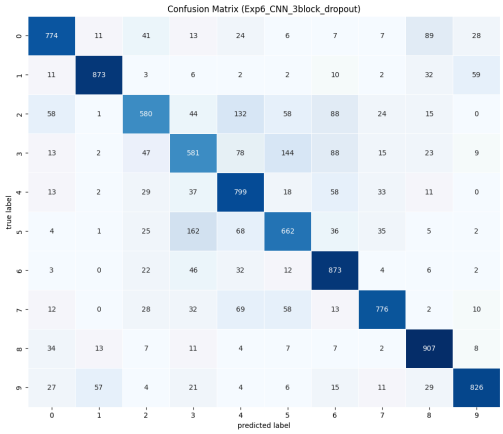



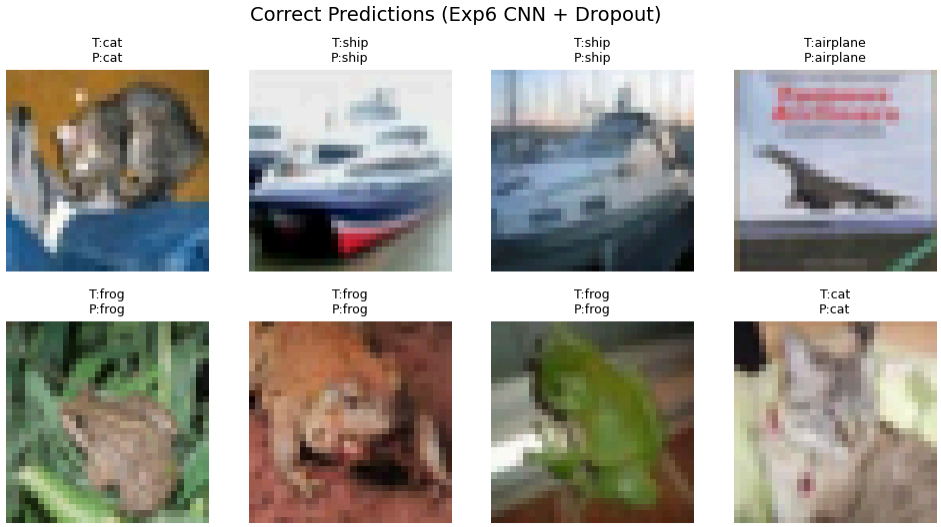





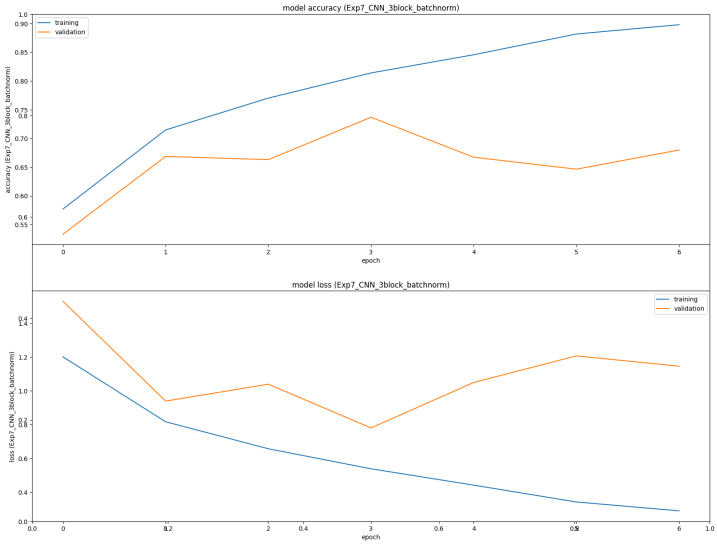

## Experiment 6 (In-Depth)

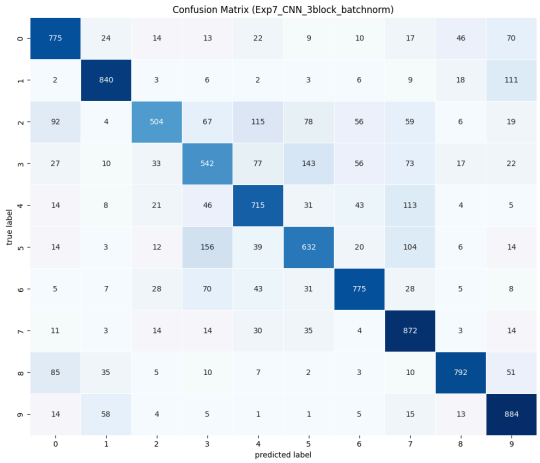


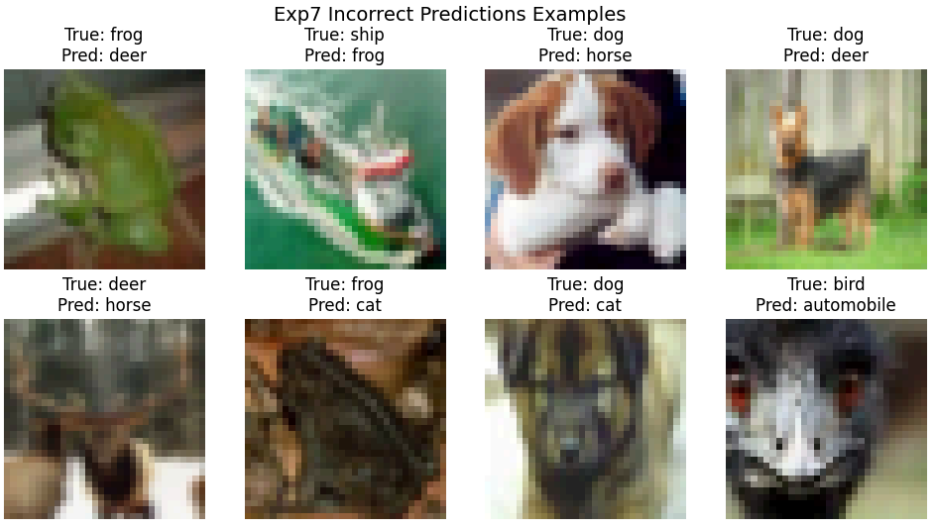


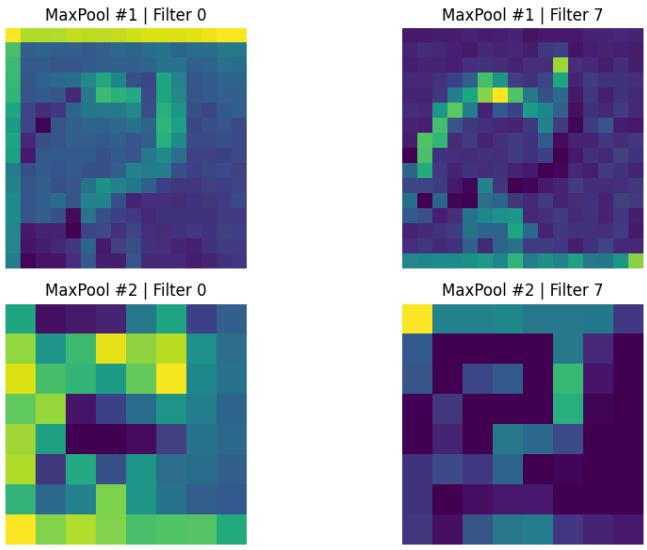

# Experiment 7 (In-Depth)

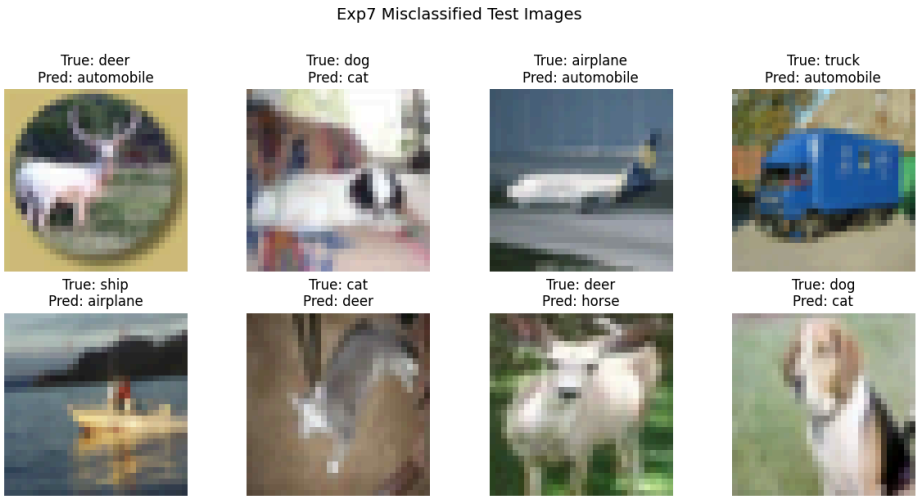



Confusion Matrix (Exp7_CNN_3block_batchnorm)
true label
predicted label
0 1 2 3 4 5 6 7 8 9
775 24 14 13 22 9 10 17 46 70
2 840 3 6 2 3 6 9 18 111
92 4 504 67 115 78 56 59 6 19
27 10 33 542 77 143 56 73 17 22
14 8 21 46 715 31 43 113 4 5
14 3 12 156 39 632 20 104 6 14
5 7 28 70 43 31 775 28 5 8
11 3 14 14 30 35 4 872 3 14
85 35 5 10 7 2 3 10 792 51
14 58 4 5 1 1 5 15 13 884

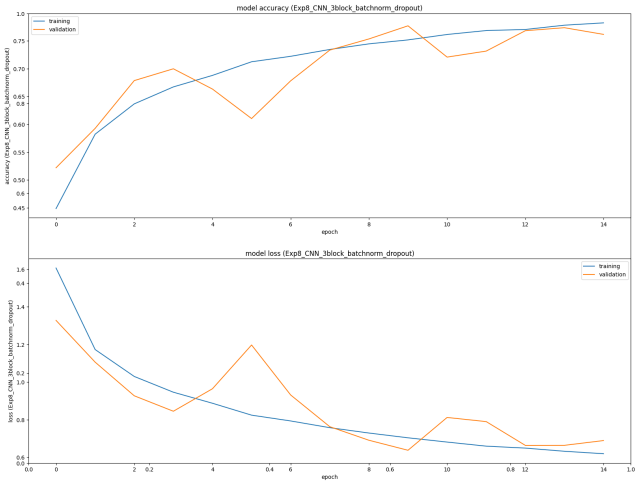

Exp7 Incorrect Predictions Examples
True: frog
Pred: deer
True: ship
Pred: frog
True: dog
Pred: horse
True: dog
Pred: deer
True: deer
Pred: horse
True: frog
Pred: cat
True: dog
Pred: cat
True: bird
Pred: automobile

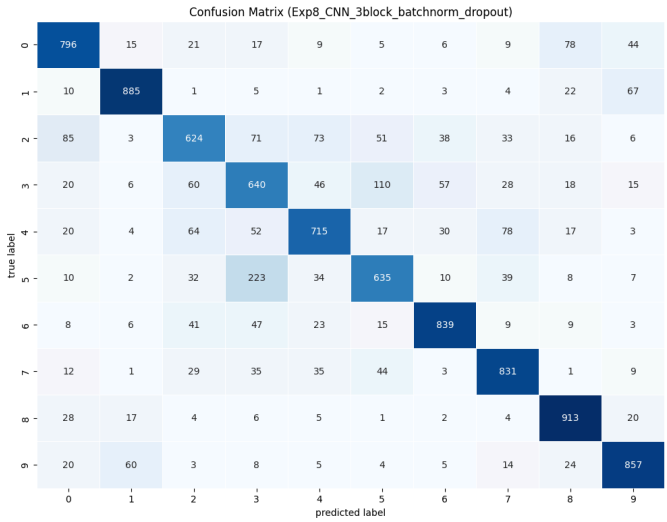
Chosen image idx=123

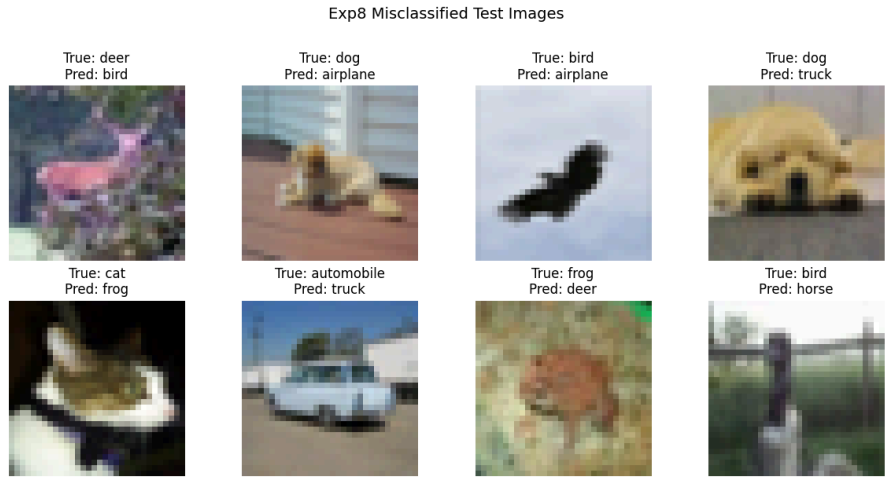


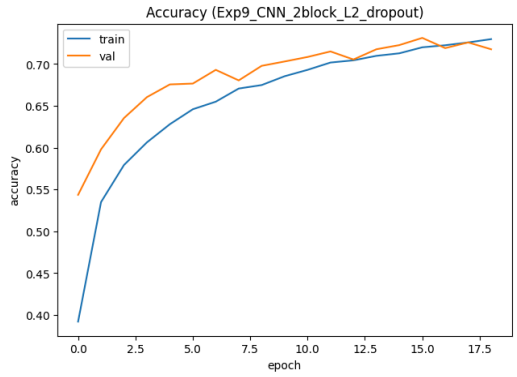

# Experiment 8 (In-Depth)

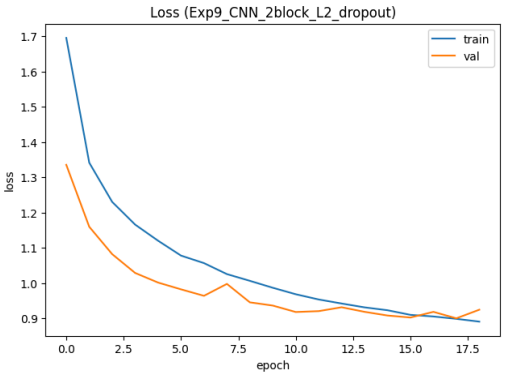


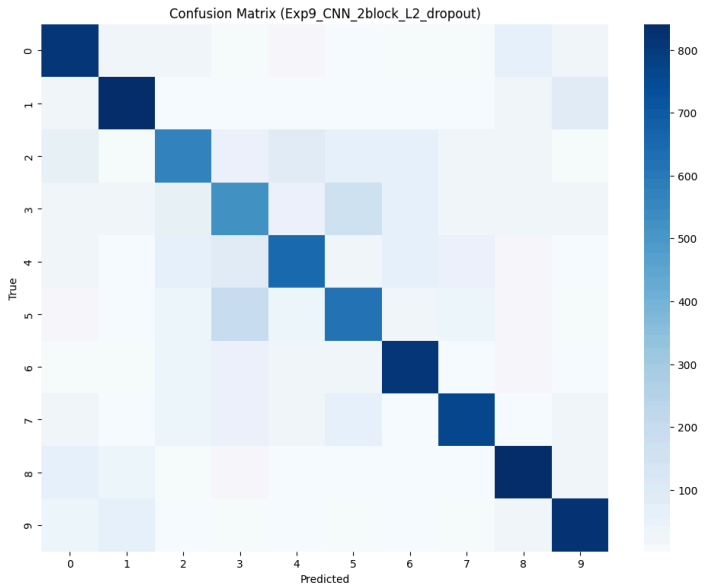


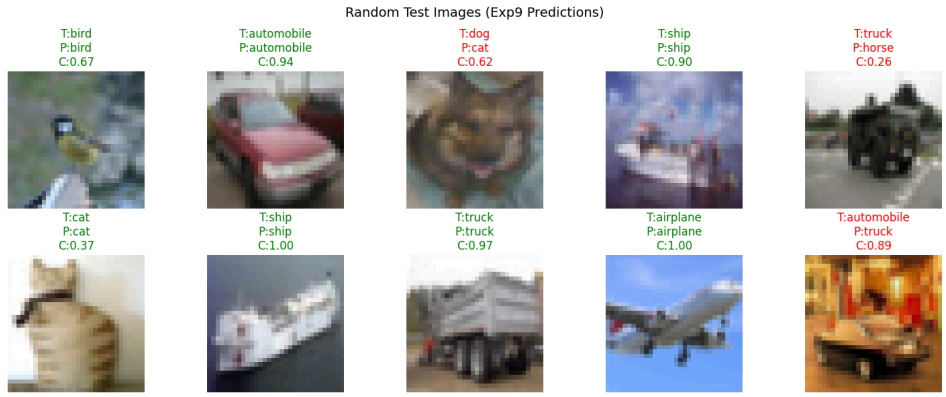



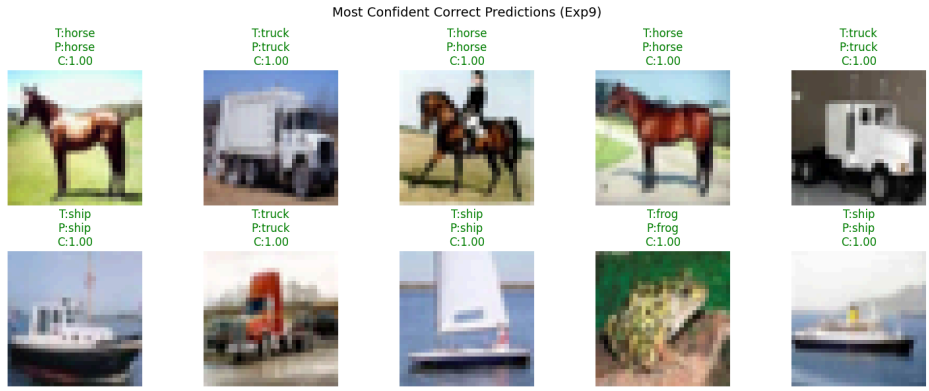

# Experiment 9 (In-Depth)

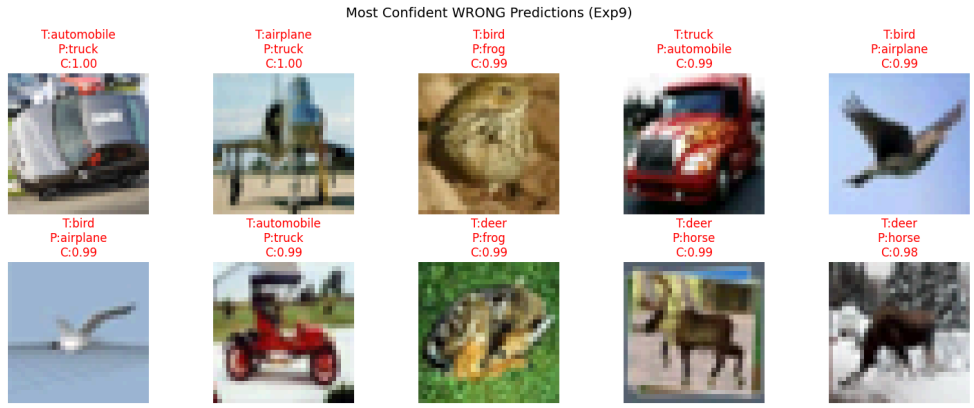


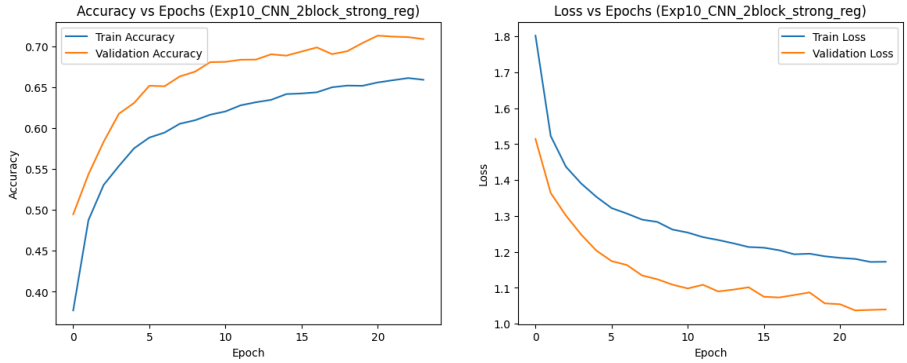

Confusion Matrix (Exp9_CNN_2block_L2_dropout)
True
Predicted
0
1
2
3
4
5
6
7
8
9
800
700
600
500
400
300
200
100

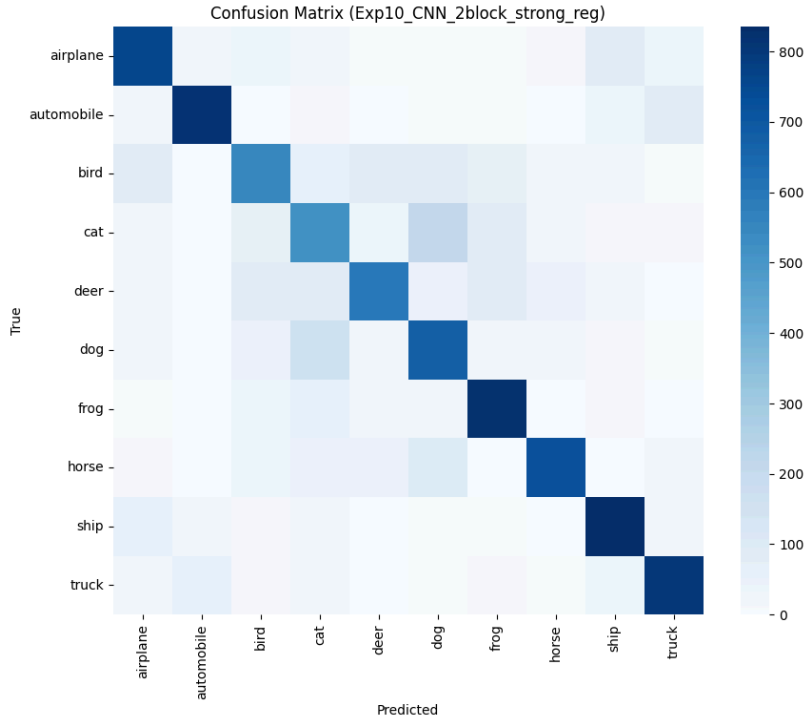

Random Test Images (Exp9 Predictions)
T:bird
P:bird
C:0.67
T:automobile
P:automobile
C:0.94
T:dog
P:cat
C:0.62
T:ship
P:ship
C:0.90
T:truck
P:horse
C:0.26
T:cat
P:cat
C:0.37
T:ship
P:ship
C:1.00
T:truck
P:truck
C:0.97
T:airplane
P:airplane
C:1.00
T:automobile
P:truck
C:0.89

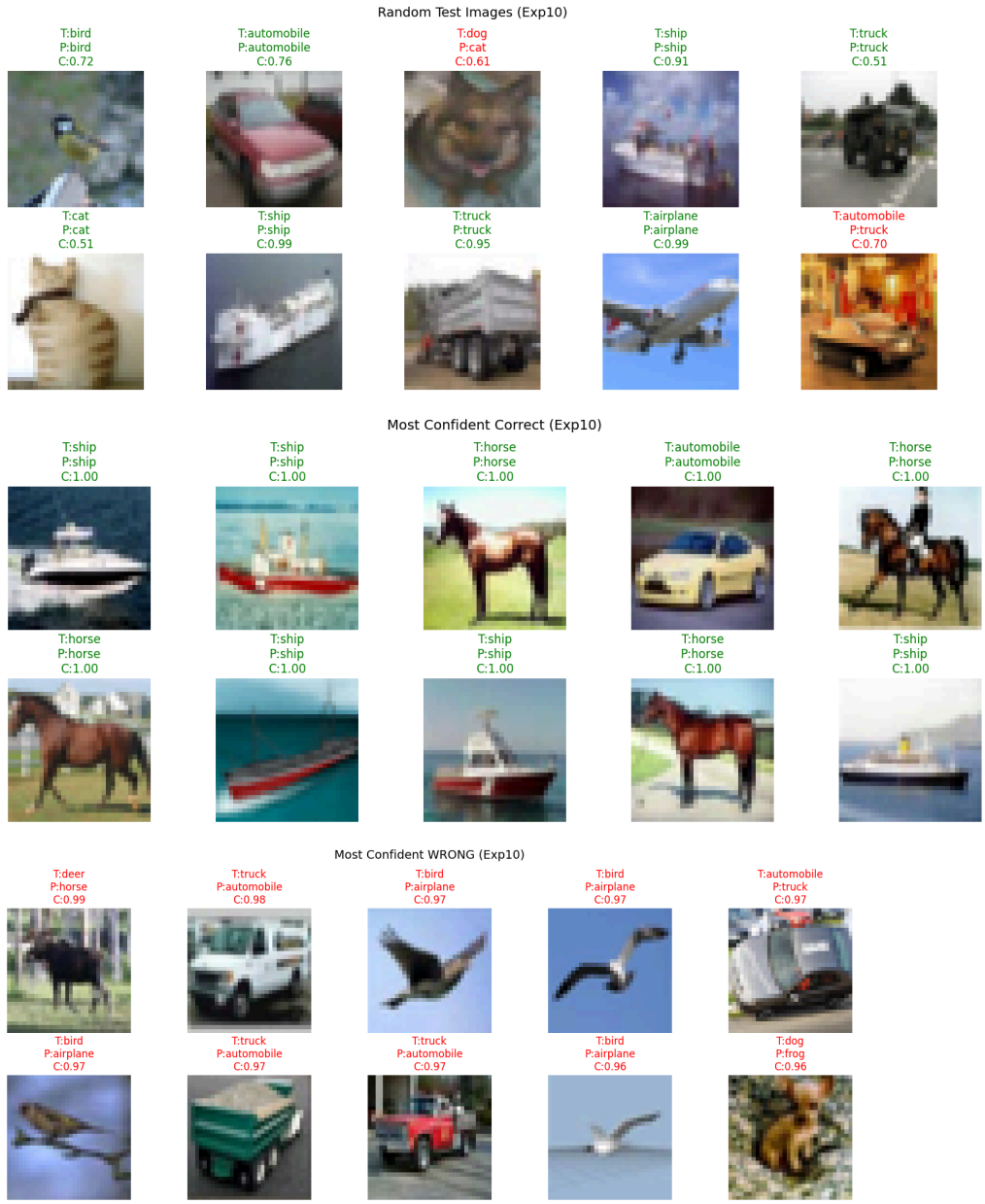

Most Confident Correct Predictions (Exp9)
T:horse
P:horse
C:1.00
T:truck
P:truck
C:1.00
T:horse
P:horse
C:1.00
T:horse
P:horse
C:1.00
T:truck
P:truck
C:1.00
T:ship
P:ship
C:1.00
T:truck
P:truck
C:1.00
T:ship
P:ship
C:1.00
T:frog
P:frog
C:1.00
T:ship
P:ship
C:1.00




# Experiment 10 (In-Depth)






# Experiment 11

| | experiment | params | train_loss | train_acc | val_loss | val_acc | test_loss | test_acc | train_time_sec | optimizer | batch_size | epochs_ran |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Exp11_DNN_3hidden_dropout_L2 | 1738890 | 1.712304 | 0.403356 | 1.750123 | 0.3886 | 1.725983 | 0.3946 | 714.75056 | adam | 64 | 20 |